

Potassium

Test Details

METHODOLOGY

Ion-selective electrode (ISE)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Evaluate electrolyte balance; followed patients on diuretic therapy and with renal diseases, particularly salt-losing nephropathy; evaluate patients being treated for acidosis; prevent cardiac arrhythmias; evaluate alcoholism with delirium tremens; evaluate and treat ketoacidosis in diabetes mellitus; evaluate acid-base balance, water balance; manage intravenous therapy; evaluate anion gap; evaluate muscular weakness, leukemia, diseases of the gastrointestinal tract including laxative abuse, large villous adenomas, emesis, fistulas and tube drainage; detect, diagnose, and manage mineral corticoid excess (primary aldosteronism, Cushing syndrome, tumor with ectopic ACTH production, some cases of congenital adrenal hyperplasia); licorice ingestion. Potassium is increased in oliguria, anuria, urinary obstruction, renal failure due to shock (decreased removal of potassium), and renal tubular acidosis. Potassium is decreased in three ways:

- Inadequate intake
- Excessive loss due to diarrhea or vomiting or decreased reabsorption due to increased secretion of mineralocorticosteroids
- Movement into the cell as occurs with conditions causing alkalosis

Custom Additional Information

Hypokalemia (low potassium) has been found in >90% of hypertensive patients with primary aldosteronism (Conn syndrome). This uncommon entity is a curable cause of hypertension. Low potassium occurs with endogenous or exogenous increase in other corticosteroids, including that in Cushing syndrome as well as with dietary or parenteral deprivation of potassium (eg, parenteral therapy without adequate potassium replacement). Hypokalemia occurs with vomiting, diarrhea, fistulas, laxatives, diuretics, burns, excessive perspiration, Bartter syndrome, some cases of alcoholism and folic acid deficiency, in alkalosis and in renal tubular acidosis as well as in other entities.

Low potassium is much more significant with a low pH than with a high pH. When pH increases by 0.1, potassium decreases approximately 0.6 mmol/L. With low pH, as in ketoacidosis, as therapeutic adjustment towards normal is made, plasma/serum K^+ levels will decrease.

Phosphorus levels tend to follow potassium levels downwards during therapy of diabetic ketoacidosis; both are largely intracellular. With insulin therapy (and increased utilization of carbohydrate), potassium moves into cells and serum/plasma level falls. Hyperalimentation may have a similar effect. Hypokalemia has been reported in slightly over one-half of a series of 32 patients with acute myelogenous leukemia,¹ but thrombocytosis can increase serum potassium levels, vide supra.

Thiazide/chlorthalidone therapy may cause hyperuricemia and hypercalcemia as well as hypokalemia.

The watery diarrhea-hypokalemia-achlorhydria (WDHA) syndrome most often is related to vasoactive intestinal polypeptide (VIP).

Hyperkalemia (high potassium) reflects generally inadequate renal excretion, mobilization of potassium from the tissues, or excessive intake or administration. Hyperkalemia occurs with hemolysis, trauma, with administration of potassium salts of some drugs, Addison disease, acidosis, insulin lack, with increased osmolality (eg, glucose, mannitol), and in other entities as well as with renal diseases. Increased potassium can occur with potassium sparing diuretics, nonsteroidal anti-inflammatory drugs, especially in the presence of renal disease. Systemic heparin therapy can suppress aldosterone release and increase potassium, especially in the presence of other factors.

A discussion of the relation between lactic acidosis and ketoacidosis and elevated serum potassium levels is provided in a paper by Fulop.²

Drug effects are summarized.³

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection; avoid hemolysis.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

Age	Range (mmol/L)
0 to 7 d	3.7–5.2
8 to 30 d	3.7–6.4
1 to 6 m	3.8–6.0
7 m to 1 y	3.8–5.3
>1 y	3.5–5.2

Storage Instructions

Maintain specimen at room temperature or refrigerate.

Causes for Rejection

Hemolysis; improper labeling; unspun or improperly spun specimen

Footnotes

1. Mir MA, Brabin B, Tang OT, Leyland MJ, Delamore IW. Hypokalaemia in acute myeloid leukaemia. *Ann Intern Med.* 1975; 82(1):54-57. PubMed 16553 (<http://www.ncbi.nlm.nih.gov/pubmed/16553>)
2. Fulop M. Serum potassium in lactic acidosis and ketoacidosis. *N Engl J Med.* 1979 May 10; 300(19):1087-1089 (review). PubMed 34793 (<http://www.ncbi.nlm.nih.gov/pubmed/34793>)
3. Hitz J, Trivin F. Potassium. In: Siest G, Galteau MM, eds. *Drug Effects on Laboratory Test Results Analytical Interferences and Pharmacological Effects*. Littleton, Mass: PSG Publishing Co Inc;1988: 362-374.